

TN SKILLS – PROJECT DOCUMENTATION

SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE

PHASE 7 – PROJECT DOCUMENTATION

7.1 Problem Statement

The current record access process does not provide sufficient control when different users require different levels of access. A secure mechanism is required to restrict record access based on user roles and field values such as Branch.

7.2 User Story

As an EEE branch user, I want to view only EEE branch records so that I can access the records relevant to my branch. As an authorized user, I want to perform the operations permitted by my assigned role. As an administrator, I want full access to all records so that I can manage the complete set of institution records.

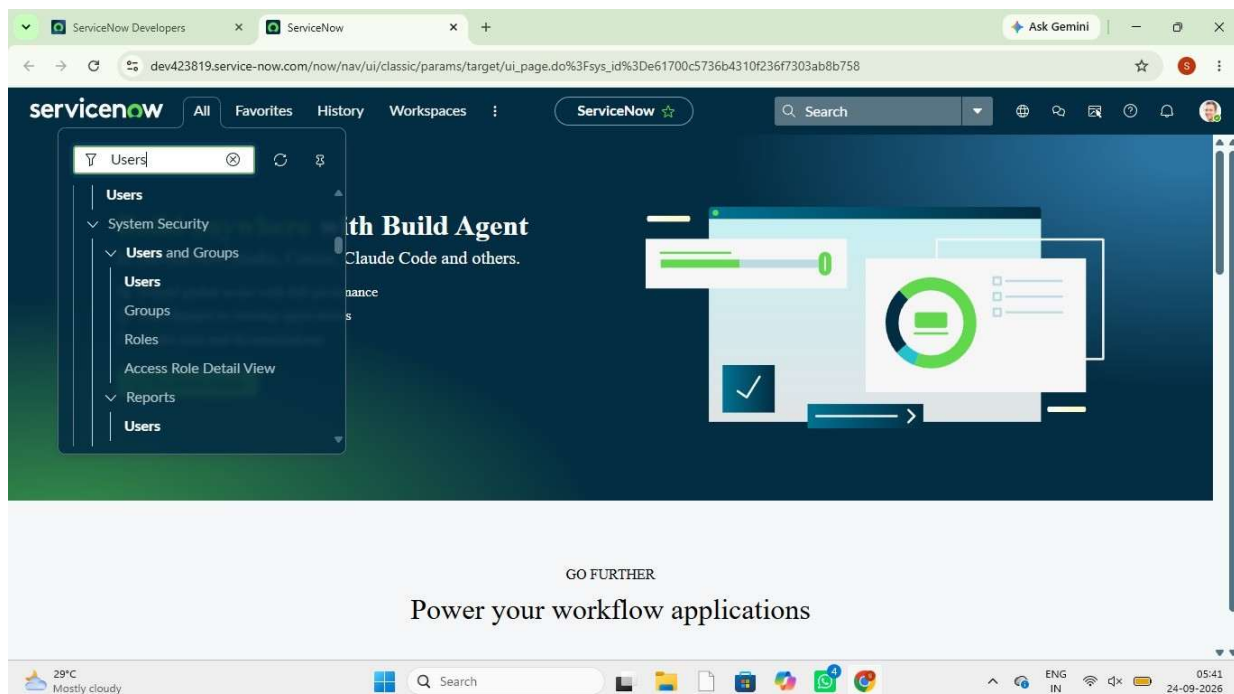
7.3 Project Objective

The objective of this project is to implement Script-Controlled Access Control Lists (ACLs) in ServiceNow to provide secure, role-based and record-level access. The project aims to restrict access based on the Branch field, control Read, Create, Write and Delete operations, provide role-based permissions, allow administrators to access all records, and improve record-level security.

7.4 TASK INITIATION

Milestone 1 : User, Role and Table Setup

Activity 1: Create the EEE User and required custom roles.



ServiceNow Developers x EEE User | User | ServiceNow x + Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D5db3bd0173a78310f236f7303ab8b7e4%26sysparm_record_target%3Dsys_use...

servicenow All Favorites History Workspaces User - EEE User Search

User EEE User Update Set Password Delete

User ID EEE User Email eeeuser@gmail.com

First name EEE Language -- None --

Last name User Calendar integration Outlook

Title Department System (America/Los Angeles)

Time zone System (2026-08-19)

Date format Business phone

Mobile phone

Photo Click to add...

Password needs reset ☐

Locked out ☐

Active ☒

Identity type Human

Internal Integration User ☐

Update Set Password Delete

Related Links

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29°C Partly cloudy Search ENG IN 05:53 24-09-2026

ServiceNow Developers

Users | ServiceNow

+

Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do

servicenow

All

Favorites

History

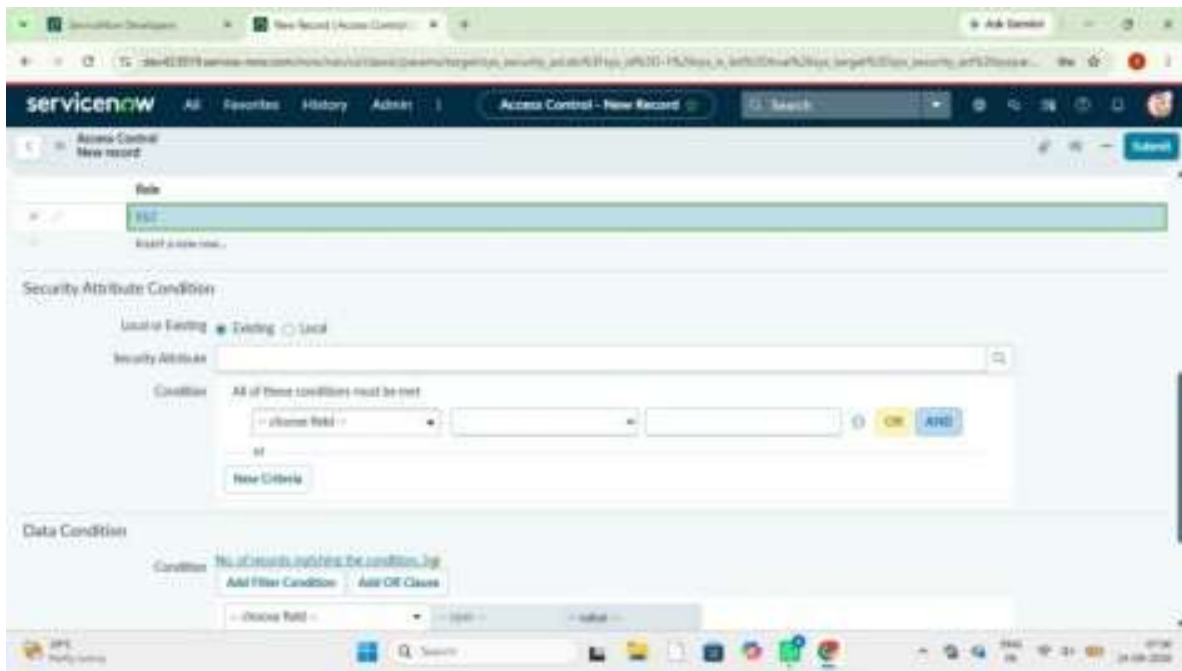
Workspaces

Users

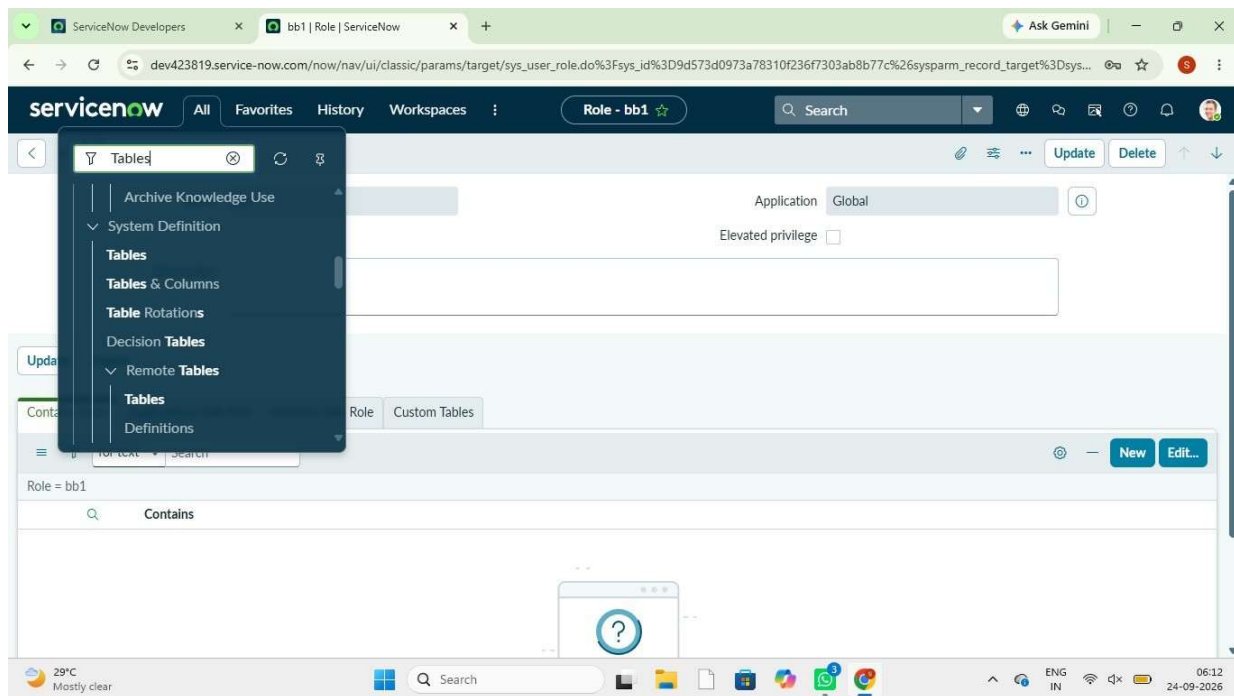
Search

Roles

</



Activity 2: Create the Institution Details custom table.



Column label	Type	Reference	Max length	Default value	Display
Updated	Integer	empty	-40	false	false
Updated	Date/Time	empty	40	false	false
Created by	String	empty	40	false	false
Faculty Name	Reference	User	32	false	false
Description	String	empty	40	false	false
Box ID	Set (1-64K)	empty	32	false	false
Student Roll Number	String	empty	40	false	false
Created	Date/Time	empty	40	false	false
Branch	Choice	empty	40	false	false
Email	String	empty	40	false	false
Updated by	String	empty	40	false	false
Student Name	Reference	User	32	false	false
Phone Number	String	empty	40	false	false

Activity 3: Add the required fields such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
Selfpageprint	Allow if	read	ui_page	true	admin	2019-02-20 01:02:07
Selfpageinsert	Allow if	read	ui_page	true	admin	2020-09-30 15:15:15
ConversationIndex	Allow if	read	ui_page	true	admin	2019-06-04 13:13:38
Sharepagepageprint	Allow if	read	ui_page	true	admin	2019-02-21 02:31:44
Spreadsheets	Allow if	read	ui_page	true	admin	2020-09-30 03:19:52
Send_email_sendEmail	Allow if	read	ui_page	true	admin	2023-05-29 05:36:27
Box_service_model_json	Allow if	read	ui_page	true	admin	2015-10-28 02:05:30
Box_service	Allow if	read	ui_page	true	admin	2014-11-04 01:14:39
Service-web-client-app	Allow if	read	ui_page	true	admin	2013-03-04 10:20:09
Get_global_search_results	Allow if	read	ui_page	true	admin	2020-06-01 00:45:57
Test	Allow if	read	ui_page	true	admin	2014-01-09 13:03:20
Spec_availability calculation_writing	Allow if	read	ui_page	true	admin	2014-02-18 06:04:39
Enable	Allow if	read	ui_page	true	admin	about 11 months ago
Support_ticket	Allow if	read	ui_page	true	admin	2020-11-01 16:44:04

Activity 4: Create sample records with different branch values such as ECE, EEE and CSE.

Screenshot of the ServiceNow web interface showing the 'Institution Details' table. The table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. The data rows are:

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
ECE			Abraham Lincoln		Abel Tuter	STU100
EEE			Adela Cervantsz		Alejandra Prenatt	STU200
CSE			Aileen Mottern		Alene Rabeck	STU300
EEE			Amos Linnan		Allie Pumphrey	STU400

The bottom of the screenshot shows a Windows taskbar with a search bar, icons for various applications, and system status information (30°C, Mostly cloudy, 20:30, 24-09-2026).

Milestone 2 : ACL Configuration

Activity 1: Create the Read ACL for the Institution Details table.

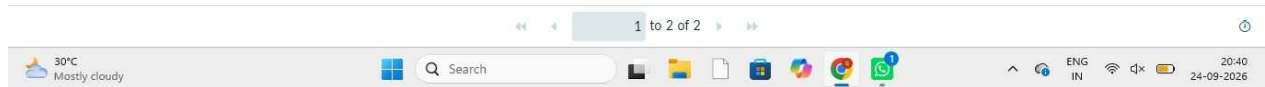
Screenshot of the ServiceNow web interface showing the 'Access Controls' table. The table has columns: Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The data rows are:

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-v-a-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$studio	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04

The bottom of the screenshot shows a Windows taskbar with a search bar, icons for various applications, and system status information (28°C, Partly sunny, 06:51, 24-09-2026).

Activity 2: Configure the Branch condition so that EEE records are controlled appropriately.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	ACL Is Executed		Adela Cervantsz		Alejandra Prenatt	STU200
EEE			Amos Linnan		Allie Pumphrey	STU400



Activity 3: Read, Create, Write and Delete ACLs for the corresponding operations.

Warning: Access, security attributes, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

Type: Record
 Operation: create
 Decision Type: Allow If
 Name: Institution Details (u_institution_details)
 Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

ServiceNow Developers

New Record | Access Control |

Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysparm_che...

servicenow

All

Favorites

History

Admin

Access Control - New Record

Search

Submit

* Type

record

Application

Global

* Operation

read

Active

☒

Admin overrides

☒

Advanced

☐

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

-- None --

Description

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

28°C

Partly sunny

Search

ENG

IN

06:55

24-09-2026

ServiceNow Developers

New Record | Access Control |

Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26syspar...

servicenow

All

Favorites

History

Admin

Access Control - New Record

Search

Submit

* Type

record

Application

Global

* Operation

write

Active

☒

Admin overrides

☒

Advanced

☐

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

-- None --

Description

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

29°C

Partly sunny

Search

ENG

IN

07:32

24-09-2026

ServiceNow Developers | New Record | Access Control | S | + | Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26syspar...

servicenow All Favorites History Admin | Access Control - New Record | Search

Access Control New record | Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record | Application: Global | Active: ☒ | Advanced: ☐

* Operation: delete | Admin overrides: ☒ | Decision Type: Allow If | Protection policy: -- None --

* Name: Institution Details [u_institution_details] | -- None -- | Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

More Info

29°C Partly sunny | Search | ENG IN | 07:37 24-09-2026

Activity 4: Configure the required roles and scripts for access control.

ServiceNow Developers | New Record | Access Control | S | + | Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_security_acl%26sysparm_che...

servicenow All Favorites History Admin | Access Control - New Record | Search

Access Control New record | Submit

Condition: No. of records matching the condition: 3 | Add Filter Condition | Add OR Clause

Branch is EEE | AND | OR | X

Advanced Condition

Controlled by References: ☐

Script: ☒ Turn on ECMAScript 2021 (ES12) mode

```

1 (function () {
2     // Allow admin users full access
3     if (gs.hasRole('admin')) { return true;
4     }
5     // Allow only EEE branch users to see EEE records
6     if (gs.hasRole('bb1')){ return true;
7     }
8     // Deny access for all others return false;
9     }());

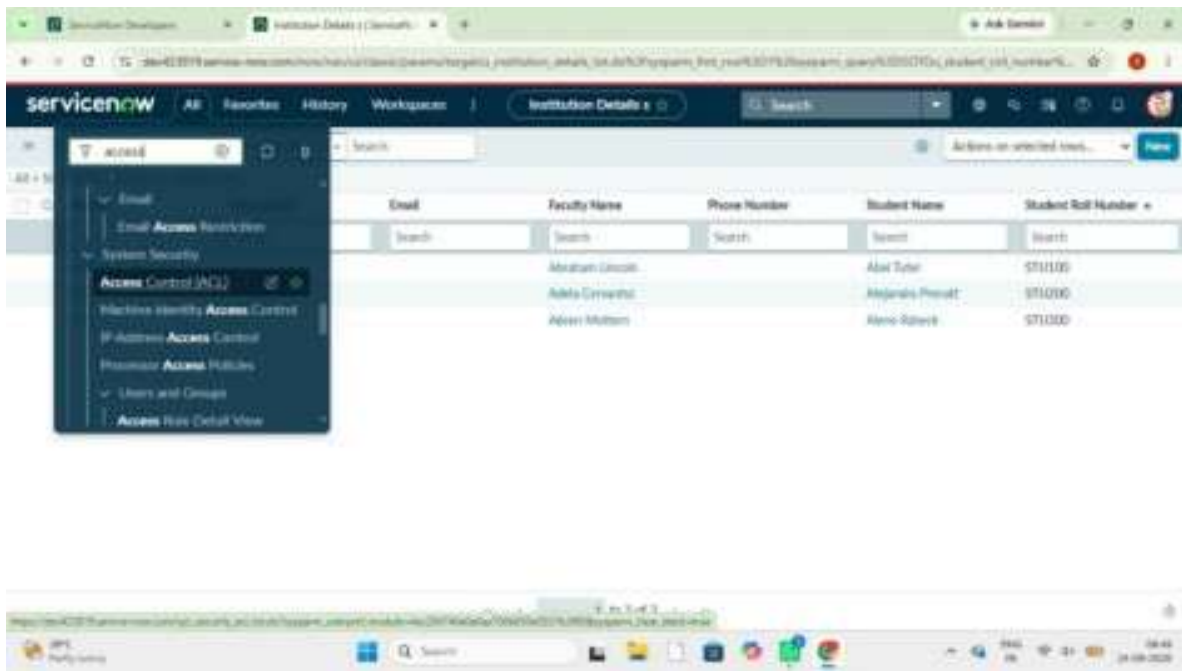
```

Could not save record because of a compile error: SyntaxError: Expected '}', got '<eof>' at line 1

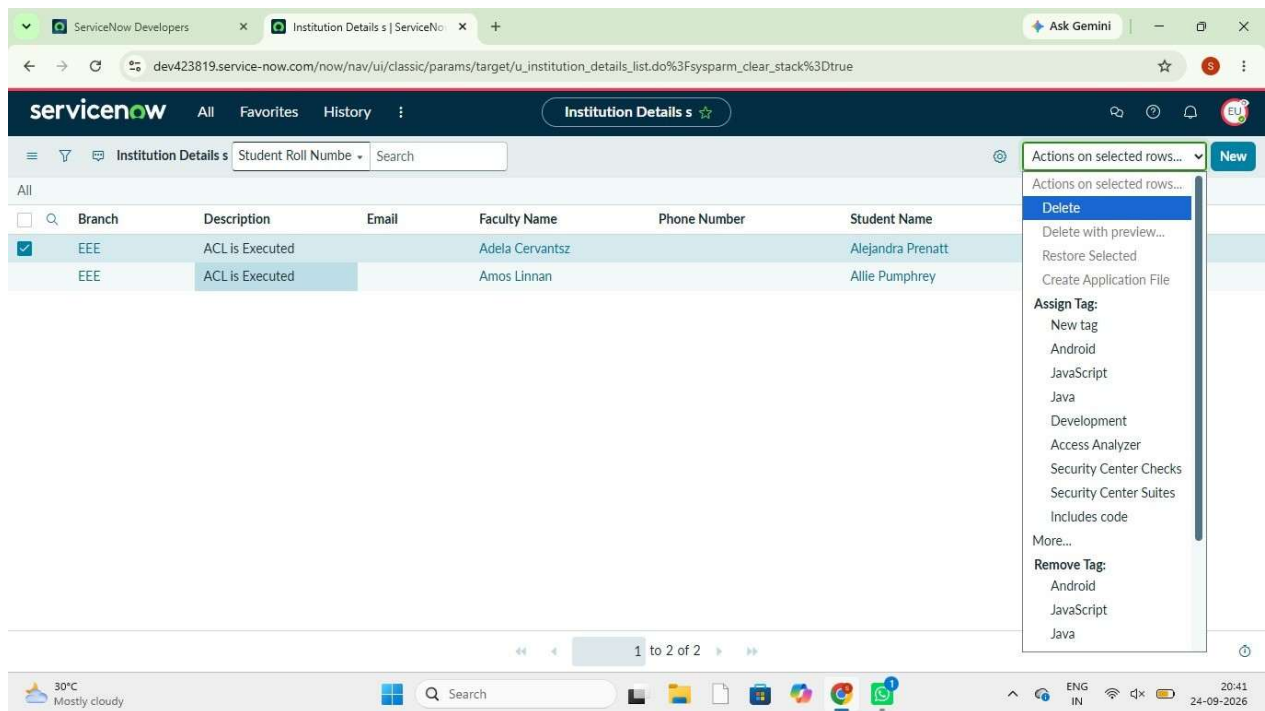
28°C Partly sunny | Search | ENG IN | 07:02 24-09-2026

Milestone 3 : Testing and Validation

Activity 1: Test the configured ACLs using the EEE



Activity 2: Verify access to EEE records and the permitted operations.



Activity 3: Verify administrator access to all records.

ServiceNow Developers | Access Controls | ServiceNow

dev423819.service-now.com/now/nav/ui/classic/params/target/sys_security_ad_list.do%3Fsysparm_userpref_module%3D4ec206740a0aa700b850e0337b2f00%26sysparm_...

servicenow All Favorites History Workspaces Access Controls Search

Access Controls Name Search Actions on selected rows... New

All

Name	Decision Type	Operation	Type	Active	Updated by	Updated
\$allappsmgmt	Allow If	read	ui_page	true	admin	2019-02-20 01:02:07
\$atf_page_inspector	Allow If	read	ui_page	true	admin	2020-03-20 15:16:15
\$conversation-builder	Allow If	read	ui_page	true	admin	2019-06-14 15:13:38
\$mycompanyappsmgmt	Allow If	read	ui_page	true	admin	2019-02-21 02:31:44
\$products	Allow If	read	ui_page	true	admin	2020-03-10 03:19:22
\$pwd_reset_serviceDesk	Allow If	read	ui_page	true	admin	2023-05-29 05:26:27
\$sa_service_model_json	Allow If	read	ui_page	true	admin	2015-10-28 02:03:33
\$sla_timeline	Allow If	read	ui_page	true	admin	2016-11-14 01:14:39
\$sn-v-a-web-client-app	Allow If	read	ui_page	true	admin	2018-03-06 10:20:09
\$sn_global_search_results	Allow If	read	ui_page	true	admin	2020-06-15 10:43:57
\$spd	Allow If	read	ui_page	true	admin	2016-01-19 12:53:23
\$spm_availability_calculation_setting	Allow If	read	ui_page	true	admin	2024-02-18 06:04:39
\$studio	Allow If	read	ui_page	true	admin	20 about 11 months ago
\$upgrade_client	Allow If	read	ui_page	true	admin	2025-11-10 16:44:04

1 to 20 of 44,052

28°C Partly sunny

Activity 4: Confirm that the configured access restrictions work as intended.

ServiceNow Developers | Institution Details | ServiceNow

dev423819.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Institution Details Search

Institution Details Student Roll Number Search Actions on selected rows... New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Adela Cervantsz		Alejandra Prenatt	STU200
EEE			Amos Linnan		Allie Pumphrey	STU400

1 to 2 of 2

30°C Mostly cloudy

ServiceNow Developers x Institution Details s | ServiceNo x + Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Institution Details s Student Roll Numb Search Actions on selected rows... New

All

	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
☐	EEE	ACL is Executed		Adela Cervantsz		Alejandra Prenatt	STU200
☐	EEE			Amos Linnan		Allie Pumphrey	STU400

1 to 2 of 2

30°C Mostly cloudy Search ENG IN 20:40 24-09-2026

ServiceNow Developers x Institution Details s | ServiceNo x + Ask Gemini

dev423819.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Institution Details s Student Roll Numb Search Actions on selected rows... New

All

	Branch	Description	Email	Faculty Name	Phone Number	Student Name
☒	EEE	ACL is Executed		Adela Cervantsz		Alejandra Prenatt
☐	EEE	ACL is Executed		Amos Linnan		Allie Pumphrey

Actions on selected rows...

Delete

Delete with preview...

Restore Selected

Create Application File

Assign Tag:

New tag

Android

JavaScript

Java

Development

Access Analyzer

Security Center Checks

Security Center Suites

Includes code

More...

Remove Tag:

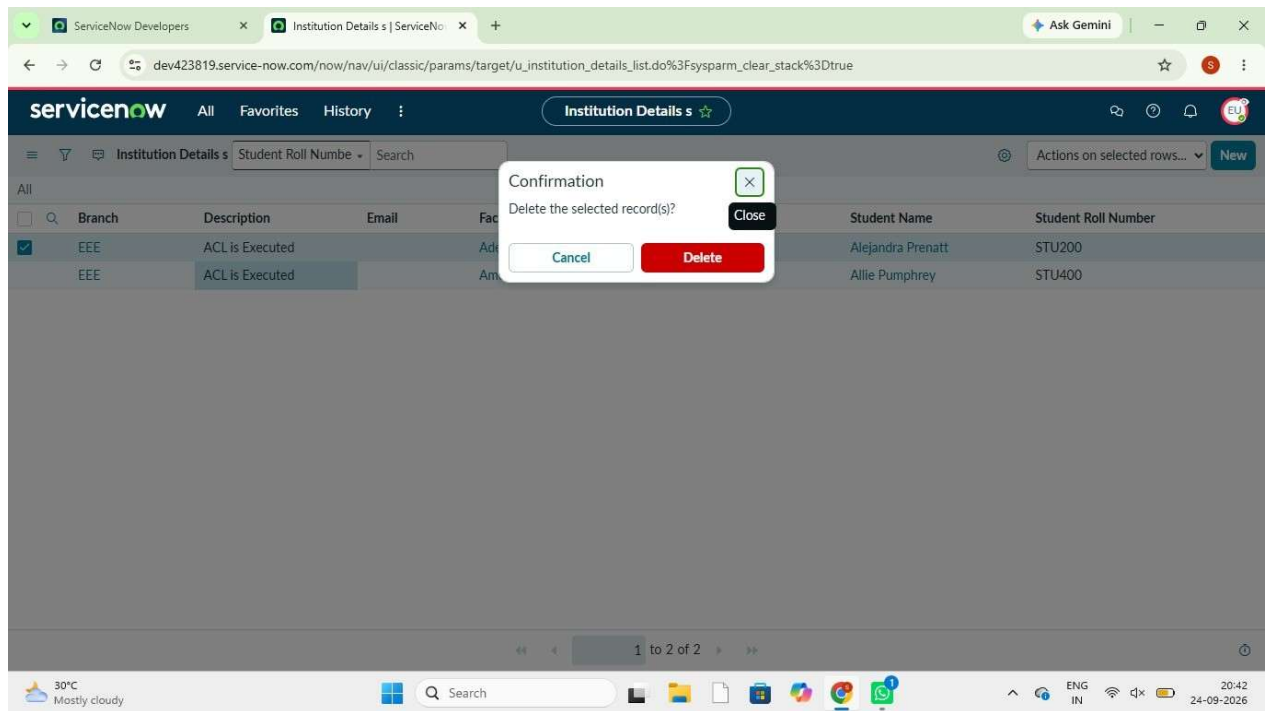
Android

JavaScript

Java

1 to 2 of 2

30°C Mostly cloudy Search ENG IN 20:41 24-09-2026



7.5 Administrator Guide

1. Open ServiceNow and access the required application.
2. Create or verify the Institution Details table and its fields.
3. Create the required custom roles and assign them to users.
4. Configure Read, Create, Write and Delete ACLs.
5. Apply the required Branch condition and script for controlled access.
6. Test the configuration using the required user and administrator accounts.
7. Verify the resulting access and record visibility.

7.6 Conclusion

The Script-Controlled ACL project provides a structured method for controlling access to ServiceNow records. By combining roles, ACL operations and field-based conditions, the system helps restrict access according to the required permissions while allowing administrators to manage the complete set of records.

7.7 Final Project Report

The final report contains the Problem Statement, User Story, Project Objectives, implementation milestones, ACL configuration steps, testing and validation details, administrator guide and conclusion. The following screenshots provide implementation evidence.

Demo Video Link

Demo Video Link: <https://drive.google.com/file/d/1PWFaUuK68bKoDZtMhoSYTkipyS1gbb/view?usp=sharing>